Sahaj Bhadja

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Summary

New-grad software engineer with 2+ years shipping **Python/TypeScript production features**, **high-throughput APIs**, and **distributed backend workflows**, with strong ownership, rapid iteration, and AI-assisted development habits.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*

- Built **production FastAPI services** and data interfaces for 500+ users, reducing latency from **1.2s to 350ms** through query optimization, schema improvements, and high-ownership iterative feature delivery.
- Implemented **asynchronous orchestration pipelines** processing 5K+ monthly jobs with retries, validation safeguards, and release checks, improving reliability while accelerating rollout of reusable backend capabilities across product workflows.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*

- Developed **React + Flask full-stack features** handling 10K+ daily requests, translating product ideas into modular APIs and user-facing workflows while shipping continuously with tight feedback loops from senior engineers.
- Leveraged AI-assisted coding workflows and CI/CD controls to improve velocity, reducing turnaround time by **60%** while maintaining quality and stability across high-impact grading and feedback product releases.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*

- Engineered **API and ETL microservice workflows** using Python/Java with AWS Glue and S3, improving response times by **30%** and supporting resilient distributed integrations for enterprise production systems.
- Maintained **20+ automated unit/integration suites** with Jenkins, PyTest, and Mockito, improving pipeline efficiency by **68%** and reducing technical debt by **20%** through rigorous release discipline.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*

- Built **Python experimentation pipelines** for 100K+ trajectory logs, creating reproducible analytics and diagnostics that enabled rapid iteration on model behavior and accelerated conversion of findings into engineering actions.
- Collaborated in high-bandwidth loops with researchers and engineers to summarize results, debug edge cases, and deliver implementation recommendations under short timelines and evolving technical requirements.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*

- Built **MERN modules** with REST APIs and WebSockets supporting eight workflows and 2M+ synchronized records, improving scalability and consistency of customer-facing operations and real-time data interactions.
- Automated Docker/Jenkins deployment and test workflows, reducing release time by **50%** and enabling faster, safer, and more repeatable shipping across iterative feature development cycles.

Skills

Languages: **Python**, **TypeScript/JavaScript**, **Java**, SQL, C++
Full Stack: React.js, Flask, FastAPI, Node.js, REST APIs, microservices
Distributed Systems: async job orchestration, high-throughput processing, reliability-first design
Cloud/DevOps: AWS (S3/Glue/RDS), Azure SQL, Docker, Kubernetes, Git, Jenkins, CI/CD
Engineering Style: AI-assisted development, rapid prototyping, code reviews, production debugging

Projects

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*

- Built a **payments-adjacent fintech platform** supporting 50K+ users and 1M+ transactions with secure APIs, real-time synchronization, and scalable deployment patterns suitable for high-volume financial product experiences.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Developed a schema-aware query workspace across 10+ databases with validated SQL generation and structured outputs, accelerating developer analysis workflows while improving reliability of data operations in production-like environments.

Linguistix | *Python, FastAPI, AWS ECS/Fargate, CodePipeline*

- Implemented a containerized API with automated validation and CI/CD release controls, achieving **87%** processing accuracy while improving deployment reliability, observability readiness, and operational consistency in cloud environments.